UNIT 3 · RAPID REVISION

Formula and Fact Sheet

Calibration (Topics 1, 6b)

- Drill: A = πDNW; rate = 10^4 × wt/A kg/ha; fluted roller rate ← exposed flute length; double-run ← gear ratio; calibrate with the actual seed lot; per-opener CV = uniformity.
- Wheel slip (5–7%) → field rate LESS than stationary calibration.
- Planter: seed spacing x = πD/(K·n); population = 10^4/(row sp × seed sp); miss >1.5x, multiple <0.5x, quality of feed 0.5–1.5x.
- Sprayer: Q(L/ha) = 600·q(L/min)/(v(km/h)·w(m)); q ∝ √p (double q → 4× p); nozzle worn if discharge >10% above rated; flat-fan = herbicide, hollow-cone = insecticide.
- 5 km/h = 30 m in 21.6 s = 1.39 m/s. 1 ha = 10,000 m².

Testing and performance evaluation (Topics 2–3)

- Tests: lab / lab-field / field; codes: RNAM (ESCAP), BIS (IS 6316 — seed-cum-fertilizer drill test), OECD (tractor). FMTTIs: Budni (central, MP), Hisar, Anantapur, Biswanath Chariali.
- Tillage quality: MWD by sieve analysis (smaller = finer tilth); penetrometer → cone index; profilometer → furrow/roughness; slip % = (1 - loaded advance/unloaded advance) × 100.
- Drill lab tests: per-row CV, hopper level & tilt effects, greased-belt spacing test, visible seed damage ≤ ~0.5%, germination after metering.
- Spreader: tray test → overlap → effective swath at CV ≤ ~15%.
- Sprayer: patternator = distribution + effective swath; VMD > NMD; dispersion ratio NMD/VMD → 1 ideal; fungicides need the highest droplet density (50–70/cm²); nozzle discharge within ±5%.

Field capacity, harvest and thresher testing, losses (Topics 4–7)

- TFC = WS/10 ha/h; η_f = EFC/TFC; combines/threshers are also rated by MATERIAL capacity (kg/s MOG, kg/h grain).
- Loss frames: pre-harvest → header → (tarpaulin) cylinder/walker/shoe; grain g/m² × 10 = kg/ha.
- Threshing eff = 100 - % unthreshed; cleaning eff at the main outlet; IS 6284 (thresher test), IS 9020 (safety), IS 9129 (chute; withdrawn, now within IS 9020).
- Chaff cutter: L = v_f/(nN); specific energy kWh/t; sprayer field η lowest (refills), rotavator/harrow highest.
- Transplanter checks: missing hills ≤ ~5%, seedlings/hill, floating hills; potato: tuber bruising; sugarcane: bud damage.

Selection, cost, matching and reliability (Topics 8–11)

- D = (C-S)/L; interest on (C+S)/2; S ≈ 10%C; tractor 10 yr/10,000 h; TIS ≈ 2%C; lube 20–30% of fuel; R&M 5–10%C/yr.
- Cost/ha falls with annual use (fixed cost spreads); A_be = FC/(custom rate - own VC).
- W = pull/(unit draft × depth); load the tractor to ~75–85%; gear-up throttle-down for light loads.
- R = e^(-t/MTBF); at t = MTBF, R = 36.8%; series R multiplies; availability = MTBF/(MTBF + MTTR).

- Timeliness sizing: required EFC = area/(days × h/day); animal costs continue on idle days.

Engine power, motors, batteries and standards (Topics 12–13)

- IP = p_m·L·A·n·K/60,000; BP = 2πNT/60,000; FP = IP − BP; η_m = BP/IP (75–85%); n = rpm/2 for 4-stroke.
- Morse test → per-cylinder IP by cut-out; Willan's line (diesel) → FP from the fuel-line intercept; motoring & retardation tests.
- η_bt = 3600/(SFC × CV); diesel SFC ≈ 250 g/kWh → ~34%.
- Ns = 120f/p; slip 3–5%; breakdown torque 2–2.5× rated; squirrel cage = farm standard; capacitor-start for high starting torque on 1-φ.
- Battery: 2.1 V/cell lead-acid, SG 1.26–1.28 charged (hydrometer), Ah capacity; Li-ion 150–250 Wh/kg + BMS.
- ISO TC 23; ISO 730 hitch; ISO 500 PTO; OECD tractor codes; ASABE D497; BIS: ISI 1947 → BIS Act 1986/2016; RNAM → ANTAM.

Further reading

- Kepner, Bainer & Barger — Principles of Farm Machinery: Ch. 2, 11, 17, 21.
- Hunt — Farm Power and Machinery Management; Liljedahl et al. — Tractors and their Power Units.
- RNAM/ANTAM (UN-ESCAP) test codes; BIS: IS 6316 (drills), IS 6284 (threshers), IS 9020 (thresher safety), IS 4468 (3-point linkage), IS 4931 (PTO).
- ISO TC 23 standards (ISO 730, ISO 500, ISO 11783); OECD tractor test codes; ASABE D497.
- FMTTI test procedures (Budni); NPTEL lectures on farm machinery, farm power and IC engines.
